Shyamalan Kannan

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Education

Seattle University — Seattle, WA
M.S. Computer Science, Data Science — *Sep 2024 – Mar 2026*

Presidency University — Bengaluru, India
B.Tech Computer Science, AI & ML — *Mar 2020 – Jun 2024*

Experience

Software Engineer — Apr 2025 – Mar 2026
Seattle University — *Seattle, WA*

- Designed CI/CD pipelines for C# applications with automated security scanning that caught vulnerabilities before deployment, cutting release cycles 20%
- Rebuilt delivery on AWS, TypeScript, and shell automation, dropping release time from weeks to hours and ending repeat configuration failures
- Worked with research, IT, and operations to set deployment strategy, and wrote 12+ infrastructure runbooks so releases repeat without tribal knowledge

Software Engineer Intern — Jul 2023 – Oct 2023
Finkey Consultancy Ltd — *Bengaluru, India*

- Built a Python/FastAPI financial data pipeline on AWS Glue, Athena, Redshift, and S3 handling 50K+ records from 15+ sources, improving retrieval 45% and backing risk analysis on $30K of capital exposure
- Trained PyTorch anomaly detection and volatility forecasting models, then wired OpenAI API summaries into real-time risk classification, cutting analyst review time 40%
- Wrote the REST endpoints and ETL orchestration that ingest, transform, and query structured financial data across distributed storage

Projects

Ozark | *Python, React, Vite, SQLite WAL, Agent Evaluation* | ozark-iota.vercel.app — May 2026

- Built a local-first agent simulation lab with runtime guardrails, adversarial scenario testing, and deployment-readiness scoring across safety, task completion, security, latency, and guardrail coverage)
- Wrote the Python evaluation framework for trace replay and diffing, guardrail enforcement, and live HTTP/stdio endpoint tests, so teams score agents before production without cloud dependencies

Bread | *Rust, Tree-sitter, ripgrep* | bread-lilac.vercel.app/docs — Jul 2026

- Built Rust middleware for coding agents that bounds oversized reads and searches, tags context with provenance, and blocks stale edits, cutting model-visible tool output 94.3% and input tokens 37.8%
- Indexed the workspace with Tree-sitter across 115 languages and ripgrep-compatible search; native extension dispatch took lookups from 4.477s to 25.431ms)

Oh My Fx | *Zig, LLM Agents, MCP, Multi-Provider Inference, CLI* | github.com/Yabuku-xD/omfx — Aug 2026

- Built a coding agent CLI in Zig shipping as a 3 MB binary with 9 model providers, 27 built-in tools, and a parse gate across 36 languages that reverts broken edits before the model sees a success
- Designed the context layer and permission model: compaction that keeps dropped turns as inspectable citations rather than an LLM rewrite, and a deterministic allowlist DSL that blocks untrusted tool output by default

Technical Skills

Languages: Python, Rust, Zig, TypeScript, SQL, C#, Bash
Frontend: React, Vite, Next.js
Backend and APIs: FastAPI, REST APIs, ETL Pipelines, OAuth, Distributed Systems
AI and Agent Systems: LLM Agents, Agent Evaluation, Guardrails, Adversarial Testing, Trace Replay/Diffing, Context Engineering, MCP, Anthropic and OpenAI SDKs, PyTorch
Data and Storage: AWS Glue, Athena, Redshift, S3, SQLite WAL
Infrastructure: AWS, CI/CD, Vercel, Trace Analysis
Developer Tooling: Tree-sitter, ripgrep, LSP, Git